

MOHAMED HELMY

Android & Kotlin Multiplatform Engineer

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Summary

Android and Kotlin Multiplatform engineer with hands-on experience shipping cross-platform apps and libraries. Specialized in Jetpack Compose, modular architecture, offline-first sync, and secure mobile systems. Proven ability to modernize legacy codebases and deliver production-ready solutions.

WORK EXPERIENCE

Android Software Engineer, *Areeb Technology*

07/2025 – Present
Cairo, Egypt

- Led KMP research and validated shared logic with native UI integration under Clean Architecture, building a production-ready cross-platform blueprint.
- Refactored a monolithic codebase into 24 modules, cutting build time 30% and allowing independent team releases.
- Modernized engineering foundations by migrating to Gradle Version Catalogs and adopting Navigation 3, cutting build times by 15%.
- Replaced NDK/JNI cryptography with Google Tink and redesigned encrypted DataStore, strengthening security.
- Migrated networking from Retrofit to Ktor, reducing API latency by 20% and adding type-safe localization for faster feature rollout.
- Owned critical Boxooo and Cadda user flows, including onboarding, KYC, signup, biometrics, password recovery, notifications, and RTL fixes.

Android Software Engineer,

02/2025 – 07/2025
Remote

Freelance | POS & ERP Platform (Kotlin Multiplatform)

- Built a production-ready offline-first POS and ERP platform using Kotlin Multiplatform for Android tablets and Desktop.
- Designed Room + Supabase synchronization to support reliable data flow during connectivity drops.
- Implemented secure RPC-based stock transfer and inventory workflows to maintain data consistency.

EDUCATION

B.Sc. in Computer Science, *Benha University*

09/2019 – 06/2023
GPA: 3.24

Researched and engineered a machine learning solution to predict diabetes, analyzing health metrics datasets and evaluating classification models.

SKILLS

Mobile & Cross-Platform

Kotlin, Jetpack Compose, Kotlin Multiplatform, Compose Multiplatform, Clean Architecture, MVVM, MVI, Coroutines, Flow, Navigation 3, Hilt, WorkManager.

Platform, Backend & Security

Ktor, Koin, Room, Swift, SwiftUI, Xcode, SPM, Firebase, Supabase, REST, GraphQL, Google Tink, DataStore.

Tools & Testing

Git, GitHub, GitHub Actions, CI/CD, Gradle, Gradle Version Catalogs, JUnit, Mockito/Mockkery, Compose UI Testing, Screenshot Testing, Maestro.

PROJECTS

Safar, *Delivery Operations Platform (Kotlin Multiplatform)*

- Built a scalable logistics platform for Android and Desktop using Compose Multiplatform and MVI.
- Developed Supabase Edge Functions for payroll logic and implemented Row-Level Security for access control.
- Designed the platform to reduce manual operational overhead by 90% through automated invoice tracking and real-time driver management.

hijri-date-picker , *Kotlin Multiplatform Library*

- Developed a production-ready KMP Hijri calendar picker supporting Android, iOS, and Desktop.
- Shipped a reusable cross-platform library with platform-consistent behavior and production use in mind.

CrossCart

- Built a single-codebase app for Android, iOS, and Desktop using Kotlin Multiplatform.
- Implemented clean MVI architecture and real-time data synchronization with Supabase.

CERTIFICATIONS

Building Industry-Level Compose Multiplatform Apps: PL-Coding | 2026

Android Developer Track: Digital Egypt Builders Initiative (DEBI) | 2025

Android Application Developer Track: ITI | 2023

Advanced Android Nanodegree: Udacity & FWD | 2022

Machine Learning Specialization: DeepLearning.AI (Coursera) | 2022

CS50: Harvard University | 2021

Algorithmic Toolbox: Coursera | 2021